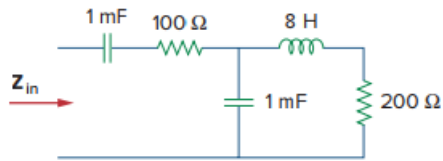


Problem

Determine the input impedance of the circuit in Fig. 9.24 at $\omega = 20$ rad/s.

Figure 9.24:



Step-by-step solution

Step 1 of 3

Refer to Figure 9.24 in the textbook.

The value of angular frequency is, $\omega = 20$ rad/sec.

Assume that,

Z_1 is the impedance of the 1 mF capacitor

Z_2 is the impedance of 8 H inductor in series with 200 Ω resistor

Z_3 is the impedance of the 1 mF capacitor in series with 100 Ω resistor

Calculate the capacitive reactance of 1 mF capacitor.

$$\begin{aligned} X_c &= \frac{1}{j\omega C} \\ &= \frac{1}{j(20)(1 \times 10^{-3})} \\ &= -j50 \, \Omega \end{aligned}$$

Calculate the inductive reactance of 8 H inductor.

$$\begin{aligned} X_L &= j\omega L \\ &= j(20)(8) \\ &= j160 \, \Omega \end{aligned}$$

Step 2 of 3

Write the expressions for three impedances.

$$\begin{aligned} Z_1 &= X_c \\ &= -j50 \, \Omega \\ Z_2 &= 200 + X_L \\ &= 200 + j160 \, \Omega \\ Z_3 &= 100 + X_c \\ &= 100 - j50 \, \Omega \end{aligned}$$

Step 3 of 3

In the Figure 9.24, the parallel combination of impedances Z_1 and Z_2 is in series with impedance Z_3 . Calculate the equivalent input impedance.

$$\begin{aligned} \mathbf{Z}_{in} &= Z_3 + Z_1 \parallel Z_2 \\ &= (100 - j50) + (-j50) \parallel (200 + j160) \\ &= (100 - j50) + \frac{(-j50)(200 + j160)}{(-j50) + (200 + j160)} \\ &= 100 - j50 + 9.597 - j55.28 \\ &= 109.59 - j105.28 \, \Omega \end{aligned}$$

Therefore, the input impedance of the circuit is $\boxed{109.59 - j105.28 \, \Omega}$.